Fiscal Year:	FY 2009	**Task Last Updated:**	FY 03/03/2009
PI Name:	Lathan, Corinna E. Ph.D.		
Project Title:	Mental Representation of Spatial Cues (3D-Space)		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20910	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	ILSRA 2001
Start Date:	05/02/2005	**End Date:**	09/30/2010
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	McCollum, Suzanne	**Contact Phone:**	281 483-7307
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Flight Program:	ISS		
Flight Assignment:	ISS Co-Investigator to ESA managed flight study that is in flight development phase.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clement, Gilles (CNRS (Centre national de la recherche scientifique))		
Grant/Contract No.:	NNJ05HD92A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Accurate perception and localization of objects in the environment are prerequisites for spatial orientation and reliable performance of motor tasks. Humans have mental representations of their environment based on sensory information and experience. Previous space studies carried out by our group have shown that cognitive functions such as mental rotation and the internal representation of the vertical axis of three-dimensional (3D) objects and written characters were impaired during and after spaceflight. The absence of the gravitational frame of reference during spaceflight could be responsible for these disturbances. The 3DSPACE experiment is designed to allow further identification of the role of gravity in the mental representation of 3D objects and environments. This experiment includes motor tests complemented by psychophysical measurements, in order to distinguish the effects of cognitive versus perceptual-motor changes due to microgravity exposure. See also http://www.nasa.gov/
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this study could have important implications for human performance during space flight, particularly with respect to tasks involving distance estimation, such as docking and teleoperation of robotic assets. There may also be consequences for a space habitat design if squared volumes do not look square to astronauts. Virtual reality training may be a way to train the astronauts to compensate for such altered spatial representation with the goal of developing countermeasures to alleviate any associated performance risks.
Task Progress:	During the current reporting period, the investigators have supported preflight, in-flight, and post-flight data collection for one crewmember of ISS Increment-17; provided Informed Consent Briefings to the crew of ISS Increments-18, -19, and -20; and conducted preflight baseline data collection (BDC) for several to the crew of ISS Increment-18, -19, and -20 crewmembers who have agreed to participate in this experiment while on station. Administrative tasks conducted during the current reporting period included weekly team meetings and teleconferences with the ESA PI, progress documentation and reporting, travel to and from meetings and data collection sites, and coordination of the training and BDC with NASA and ESA personnel. One ISS Increment-17 crewmember volunteered for this experiment. This crewmember launched on STS-123 and returned on STS-126. This was the first subject to perform the experiment in-flight. Preflight BDC was conducted for this subject just prior to the current reporting period. Computer entries (finger trackball mouse actions and digital writing tablet coordinates) were recorded during all the completed sessions. In-flight data was delivered to the PI on 22 December 2008. Data corresponding to all the flight sessions were identified and the quality of data was determined to be good. Three preflight BDCs have been conducted for an additional 5 crewmembers from Increments-18, -19, and -20. Preflight BDCs are scheduled for 2 additional subjects prior to the completion of the current reporting period. This is the first time an astronaut has been tested with 3D stereoscopic images during and after a long-duration space flight. The subject experienced no motion sickness symptoms throughout the testing and reported interesting changes in perceived depth and distance. Post-flight BDC sessions at R+0 and R+1 or 2 were not performed at the request of the flight surgeon and crewmember. While the complex writing and drawing task data, as well as the recently collected BDC data, are currently being processed, preliminary results have been summarized in the annual report comparing ground-based baseline data for control subjects (N=56), preflight data for subjects 1-4, and in-flight data for Subject 1.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Clement G, Lathan C, Lockerd A, Bukley A. "Mental representation of spatial cues in microgravity: Writing and drawing tests." Acta Astronautica 2009, Apr-May;64(7-8):678-81. http://dx.doi.org/10.1016/j.actaastro.2009.01.001 , Apr-2009
Articles in Peer-reviewed Journals	Clement G, Lathan C, Lockerd A. "Perception of depth in microgravity during parabolic flight." Acta Astronaut. 2008 Oct-Nov;63(7-10):828-32. http://dx.doi.org/10.1016/j.actaastro.2008.01.002 , Oct-2008